

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I BALA S (192524427) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bala

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

A company with unpredictable traffic spikes during seasonal sales can use cloud auto-scaling and load balancing to maintain high performance and availability.

- Auto-scaling automatically increases or decreases the number of servers
- During peak sales, more instances are added, and after traffic reduces, unnecessary instances are removed to save cost.
- Load balancing distributes incoming user requests evenly across all available servers

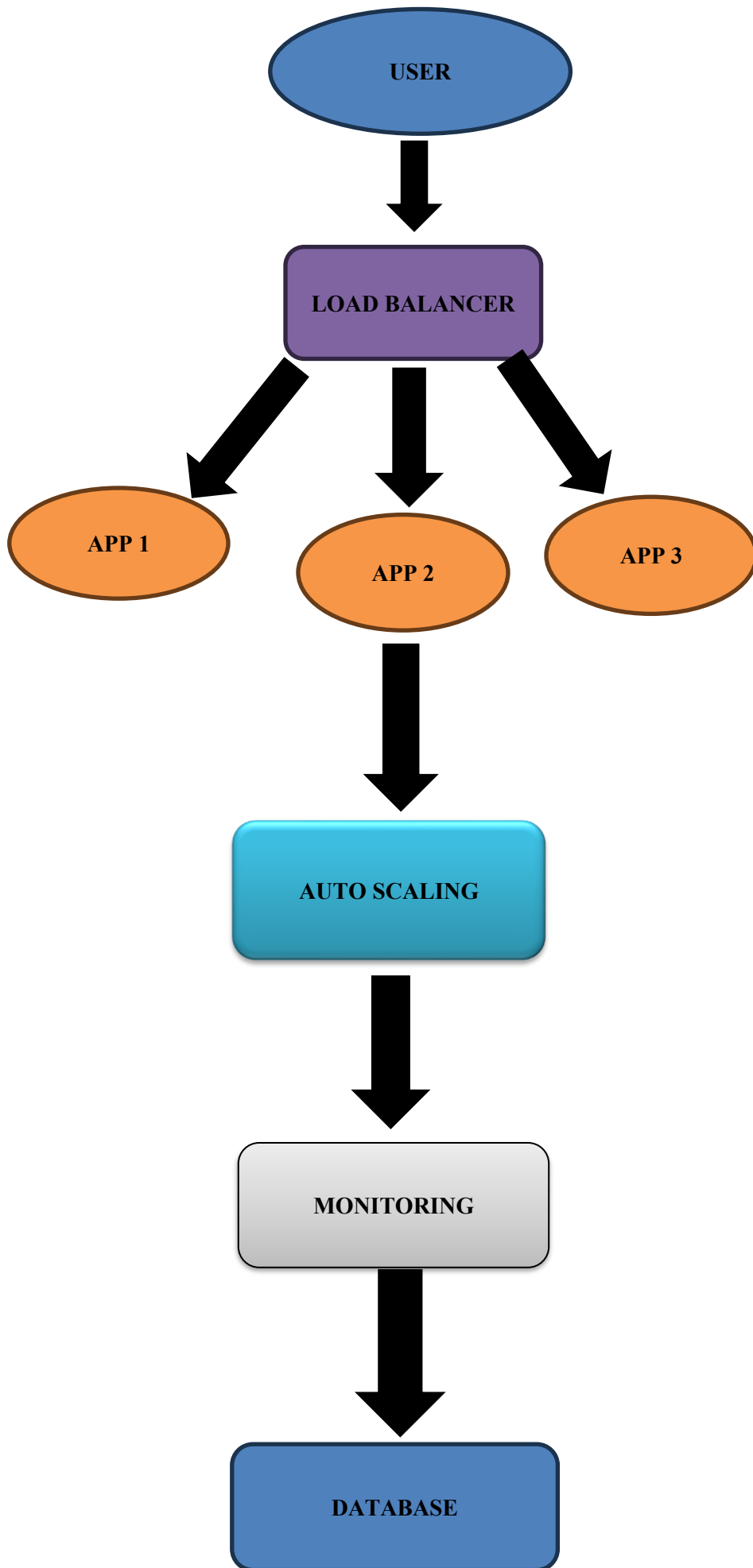
Monitoring tools:

- CPU, memory, network usage, and response time monitors continuously track system performance.
- When predefined thresholds are reached, they trigger auto-scaling actions and send alerts to administrators if any issue occurs.

Real-world example:

- During the Amazon Prime Day sale, millions of customers visit the website simultaneously.
- Cloud auto-scaling adds extra servers to handle the increased demand, while load balancing distributes traffic evenly across them.
- Monitoring tools continuously observe system health, ensuring high availability, minimal downtime, and a smooth shopping experience for customers.

1. Users send requests through the Internet.
2. DNS directs traffic to the Load Balancer.
3. The Load Balancer distributes requests across multiple web/application servers.
4. The Monitoring Tool continuously checks CPU usage, memory and network traffic
5. When traffic increases, Auto Scaling automatically launches new server instances
6. When traffic decreases, unnecessary instances are removed to reduce costs.



2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

- Organizations use cloud storage to store large volumes of backup data because it is secure, reliable, and scalable.
- Cloud providers automatically manage storage infrastructure, reducing the need for on-premises hardware.

Durability and Data Redundancy:

- Cloud storage ensures data durability by storing multiple copies of backup data across different servers and data center.
- If one server, disk, or even an entire data center fails, another copy is available.
- This redundancy protects data from hardware failures, accidental deletion, and natural disasters, ensuring high availability and reliability.

Encryption and Access Control:

Encryption:

- Backup data is encrypted while being transferred (in transit) and while stored.
- Even if unauthorized users gain access, the data remains unreadable without the encryption key.

Access Control:

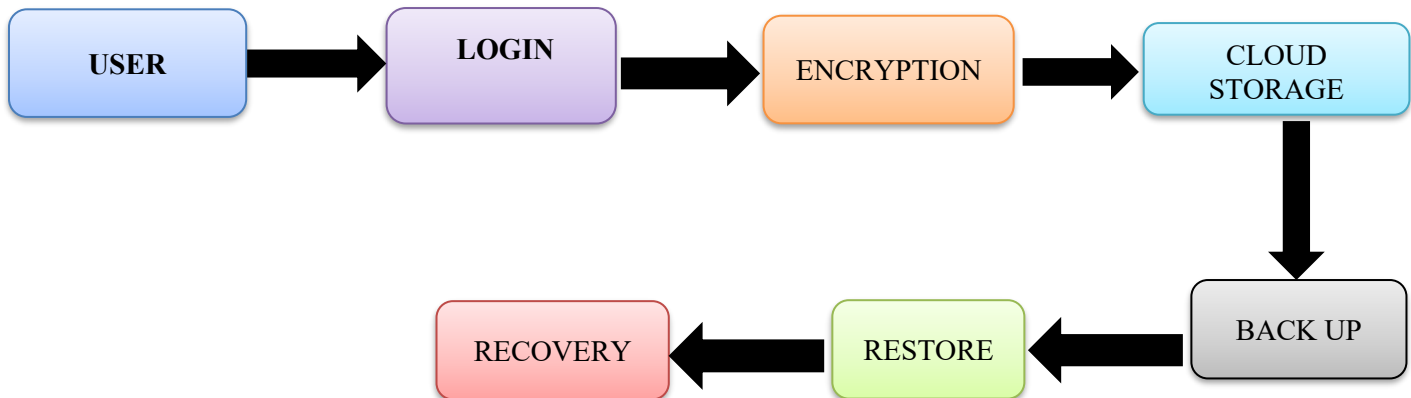
- Identity and Access Management (IAM)
- Multi-Factor Authentication (MFA)

Cost Advantages:

- No need to buy expensive storage hardware.
- No maintenance or hardware replacement costs.
- Pay only for the storage used .

Disaster Recovery Example:

- A hospital stores patient records in cloud backup storage.
- A flood damages its local data center, making on-site servers unavailable.
- Since the data is securely backed up in the cloud.
- The hospital quickly restores all patient records to new systems.



3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Global Mobile App Deployment using CDN:

- A mobile app company deploying its application globally uses a Content Delivery
- Network (CDN) to deliver content quickly and efficiently to users worldwide.
- CDNs improve performance, reduce latency, and ensure high availability.

CDN Improve Performance and Reduce Latency

- A CDN stores copies of app content on multiple edge servers located around the world.
- When a user requests content, it is delivered from the nearest edge server instead of the main server.
- This reduces the distance data travels, resulting in faster loading times and lower network latency.

Edge Servers Enhance User Experience:

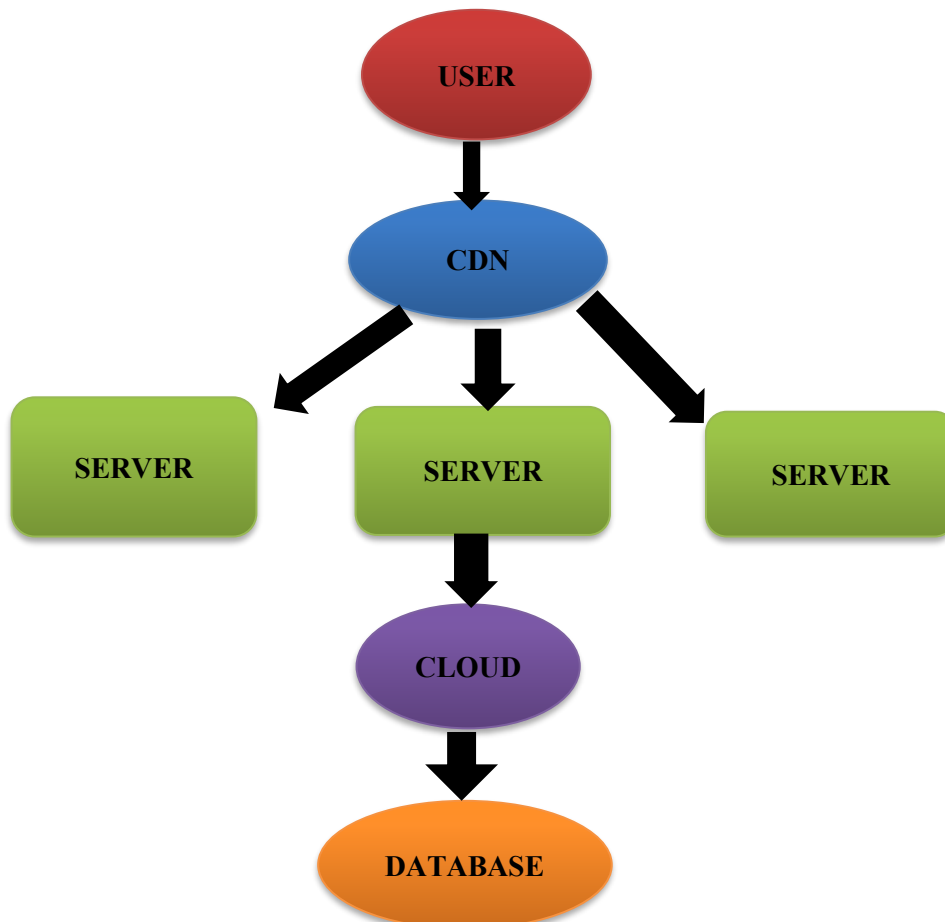
- Reduce response time.
- Minimize buffering and delays.
- Handle heavy traffic efficiently.
- Improve application speed and reliability during peak usage.

Cloud Provider Support for Global Availability:

- Cloud providers such as Amazon Web Services (AWS), Microsoft Azure.
- Google Cloud offer multiple global regions, availability zones.
- They automatically route users to the nearest healthy server, provide load balancing.
- Replicate data across regions to ensure continuous service even if one region fails.

Real-Time Streaming:

- A video streaming platform or online multiplayer game uses a CDN to deliver videos
- Players experience low latency, smooth gameplay, and minimal buffering
- Users from different countries receive the same high-performance service.



4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Regulatory Compliance and Auditing:

- Cloud providers comply with standards such as ISO 27001, PCI DSS, GDPR, and SOC 2.
- They provide audit logs and compliance reports to help organizations verify that security policies and regulations are being followed

Role of Logging, Monitoring, and Encryption:

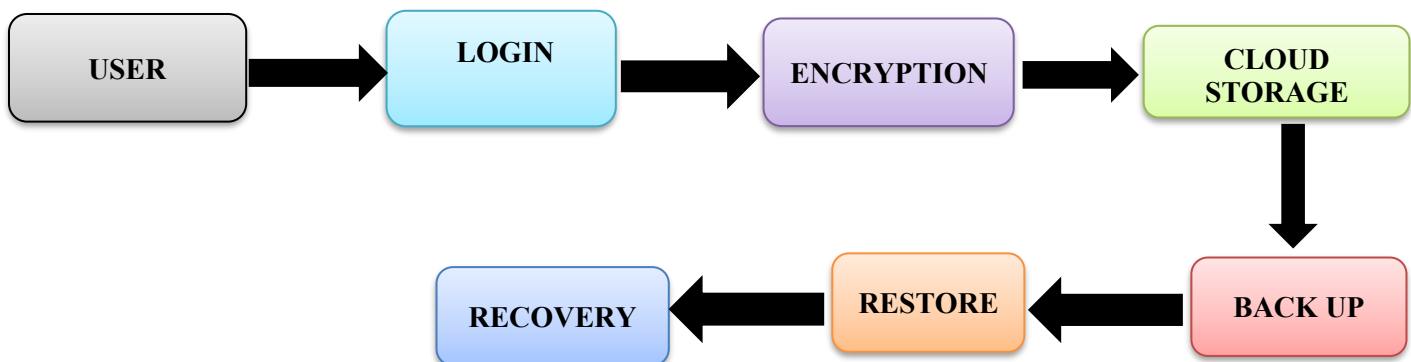
- Logging: Records all user activities and system events for auditing.
- Monitoring: Continuously detects suspicious activities and sends alerts for quick response.
- Encryption: Protects data both at rest and in transit, ensuring only authorized users can read sensitive information.

Ensuring Data Sovereignty:

- To meet data sovereignty requirements, the company stores customer data in specific cloud regions where local laws require it to remain.
- Cloud providers allow organizations to choose the geographic region for data storage and backup.

Example:

- A bank processes online fund transfers.
- Transaction data is encrypted, all access is logged and monitored, and customer records are stored only in the country's approved cloud region.
- This ensures secure financial transactions while meeting government regulations.



5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Containers vs Virtual Machines:

- Containers share the host operating system, making them lightweight, fast to start, and use fewer resources.
- Virtual Machines (VMs) include a full operating system, making them larger, slower to start, and requiring more CPU, memory, and storage

Kubernetes Orchestration:

- Automatically deploys containers.
- Scales applications based on demand.
- Restarts failed containers.
- Balances traffic across multiple containers.

Portability Across Cloud Environments:

- Containers package the application.
- Container to run on AWS, Azure, Google Cloud.
- On-premises without modification.

Example

- User Service
- Product Service
- Payment Service
- Order Service

